

YOKOHAMA, Japan

WMO: 476700

Lat: 35.4397N

Lon: 139.6525E

Elev: 43

StdP: 100.81

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	1.5	-13.6	1.2	4.8	-11.8	1.4	4.7	10.4	7.8	9.4	8.0	4.2	0	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	32.8	25.7	31.7	25.2	30.7	24.9	26.5	30.9	26.0	30.0	25.5	29.3	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	20.6	28.8	24.9	20.0	28.5	24.4	19.4	28.1	83.2	31.1	80.8	30.0	78.8	29.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.2	7.2	-0.8	34.9	0.8	1.1	-1.4	35.6	-1.9	36.3	-2.3	36.9	-2.9	37.6		
			-4.2	27.2	0.9	0.8	-4.9	27.8	-5.4	28.3	-6.0	28.7	-6.6	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.3	6.7	9.8	14.8	19.3	22.1	26.2	27.6	24.1	18.9	13.4	8.8	(6)
(7)		DBStd	7.76	2.08	2.56	3.15	3.19	2.55	2.45	2.68	2.37	3.04	2.75	2.66	2.44	(7)
(8)		HDD10.0	317	117	97	43	4	0	0	0	0	0	3	54	(8)	
(9)		HDD18.3	1566	373	325	263	112	20	2	0	1	25	147	296	(9)	
(10)		CDD10.0	2713	2	6	38	146	287	364	502	547	424	276	106	16	(10)
(11)		CDD18.3	921	0	0	0	5	48	117	244	288	175	43	1	0	(11)
(12)		CDH23.3	7085	0	0	0	9	119	469	2164	3001	1206	117	0	0	(12)
(13)		CDH26.7	2147	0	0	0	0	9	73	684	1078	291	12	0	0	(13)
(14)	Wind	WSAvg	3.5	3.6	3.7	3.9	3.9	3.6	3.2	3.4	3.4	3.3	3.3	3.4	(14)	
(15)	Precipitation	PrecAvg	1566	59	64	113	131	148	169	159	132	210	198	115	68	(15)
(16)		PrecMax	2109	138	133	204	263	297	336	409	362	465	667	282	196	(16)
(17)		PrecMin	1123	0	0	23	40	62	59	29	7	30	6	7	7	(17)
(18)		PrecStd	255	38	38	48	57	62	62	103	86	108	147	67	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.4	20.7	24.5	27.8	30.4	33.8	34.5	32.5	28.2	22.1	19.2	(19)
(20)		MCWB	11.4	13.3	13.8	16.4	19.0	23.3	25.7	26.3	24.7	21.3	17.1	14.7	(20)	
(21)		2%	DB	13.3	15.3	18.6	22.5	25.8	28.6	32.4	33.2	30.9	25.9	20.5	16.2	(21)
(22)		MCWB	8.4	10.9	13.0	15.6	18.4	22.2	25.4	26.1	24.3	20.6	15.7	11.0	(22)	
(23)		5%	DB	11.6	12.9	16.9	21.1	24.6	27.2	31.4	32.2	29.6	24.6	19.3	14.3	(23)
(24)		MCWB	6.4	8.1	12.0	15.4	18.0	21.3	25.1	25.6	23.9	19.8	14.6	9.6	(24)	
(25)		10%	DB	10.2	11.1	15.4	19.8	23.4	25.9	30.3	31.2	28.3	23.2	18.0	12.9	(25)
(26)		MCWB	5.2	6.3	10.8	14.4	17.6	21.0	24.7	25.3	23.5	18.5	13.8	8.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	14.9	16.1	20.0	21.9	24.7	26.8	27.5	26.2	23.5	19.0	16.3	(27)
(28)		MDCB	14.8	17.4	18.0	21.3	24.2	28.9	31.8	32.5	29.7	25.9	21.2	18.8	(28)	
(29)		2%	WB	9.0	11.5	14.5	17.8	20.6	23.7	26.2	26.8	25.5	22.1	17.0	12.2	(29)
(30)		MDCB	12.1	14.7	17.4	20.9	23.7	26.9	30.7	31.3	28.7	24.8	19.4	15.3	(30)	
(31)		5%	WB	7.2	8.8	13.0	16.5	19.6	22.8	25.6	26.3	24.9	20.8	15.8	10.3	(31)
(32)		MDCB	10.7	12.0	16.2	19.7	22.8	25.9	29.8	30.4	28.1	23.6	18.3	13.4	(32)	
(33)		10%	WB	5.9	7.1	11.2	15.4	18.8	21.9	25.1	25.9	24.4	19.4	14.5	9.0	(33)
(34)		MDCB	9.6	10.4	14.8	18.6	22.0	24.7	29.0	29.7	27.5	22.5	17.1	12.3	(34)	
(35)	Mean Daily Temperature Range		MDBR	6.9	6.9	7.3	7.3	6.8	5.8	5.8	6.0	5.6	5.8	6.3	6.6	(35)
(36)		5% DB	MCDBR	8.8	9.3	9.1	8.6	8.3	7.5	7.2	7.1	6.7	7.2	7.8	8.6	(36)
(37)		MCWBR	6.3	7.2	6.6	5.1	4.2	3.4	2.8	2.6	3.0	4.3	5.6	6.3	(37)	
(38)		5% WB	MCDBR	8.3	8.9	8.4	7.3	7.2	6.2	6.8	6.7	6.1	6.5	7.0	8.0	(38)
(39)		MCWBR	6.8	7.6	7.2	5.2	4.5	3.2	2.8	2.7	3.0	4.6	6.1	6.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.381	0.430	0.494	0.543	0.565	0.594	0.586	0.574	0.513	0.485	0.427	0.383	(40)	
(41)		taud	2.248	2.107	1.949	1.874	1.880	1.878	1.943	1.964	2.104	2.105	2.184	2.261	(41)	
(42)		Ebn at Noon	796	793	773	758	747	723	726	725	750	729	735	766	(42)	
(43)		Edh at Noon	107	138	175	198	200	200	186	179	149	137	114	100	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.80	3.29	4.10	4.79	5.14	4.50	4.97	5.01	3.87	3.04	2.52	2.45	(44)	
(45)		RadStd	0.20	0.34	0.27	0.48	0.55	0.43	0.85	0.46	0.35	0.33	0.24	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page